

Review Analysis Project Recap

Scope

- Goal: open-ended review analysis that discovers evaluable entities from text and ranks them by aggregated sentiment.
- Data: reviews in reviews.csv; outputs saved to entity_stats.csv.

Environment setup

- GitHub CLI was missing; installed locally at /home/lbk/.local/bin/gh. Repo cloned via git.
- Poetry installed and used for dependency management.
- Poetry keyring disabled to avoid DBus errors: poetry config keyring.enabled false.
- pyproject.toml updated with deps for notebook (jupyterlab, ipykernel, pandas, numpy, transformers, sentence-transformers, scikit-learn, yake, torch).

Notebook changes (entity_sentiment_discovery.ipynb)

- Entity extraction is open-ended (no fixed categories).
- Added NER-based extraction using Davlan/bert-base-multilingual-cased-ner-hrl.
- Added regex extraction for numeric entities (e.g., $\frac{1}{51}$, $\frac{51}{1}$).
- YAKE still used for keyphrases; all sources are merged and lightly filtered.
- Low-confidence NER mentions are tagged as entity_unknown instead of dropped.
- Sentiment is computed at the entity-mention level using the sentence that contains the entity.
- Entities are clustered by embedding similarity for grouping, but ranking is per-entity.
- entity_stats.csv now includes entity_tag and entity_cluster columns.

Issues addressed

- $\frac{1}{51}$, $\frac{51}{1}$ missing: normalized extraction and added recall improvements (now open-ended).
- Meaningless clusters: removed forced category clustering; cluster is just for grouping.
- Needed entity-level sentiment: now uses sentence-level context per mention.
- VS Code showing old notebook: reopen file, restart kernel, select Poetry interpreter.
- Syntax error in cell 5 fixed (regex string for sentence split).

How to run

- poetry shell
- Open entity_sentiment_discovery.ipynb in VS Code, pick the Poetry kernel, Restart + Run All.

Open tasks / knobs

- Adjust NER confidence threshold (default 0.70) if recall/precision needs tuning.
- Tune YAKE top-k or filtering if too much noise or missing entities.